like water quality, aeration, regular feeding, observation for mortality should be checked. Parasitic infestations and microbial infections should be observed periodically.

Economic importance of fish

Fishes form a rich source of protein food and provide a good staple food to tide over the nutritional needs of man. Fish species such as sardines, mackerel, tuna, herrings have high amino acids concentrations particularly histidine which is responsible for the meaty flavor of the flesh. It is rich in fat such as omega 3 fatty acids. Minerals such as calcium, magnesium, phosphorus, potassium, iron, manganese, iodine and copper. Some of the fish by - products are:

Fish oil is the most important fish by product. It is derived from fish liver and from the fish body. Fish liver oil is derived from the liver which is rich in vitamin A and D, whereas fish body oil has high content of iodine, not suitable for human consumption, but is used in the manufacture of laundry soaps, paints and cosmetics.

Fish meal is prepared from fish waste after extracting oil from the fish. The dried wastes are used to prepare food for pig, poultry and cattle. The wastes obtained during the preparation of fish meal are widely used as manure.

Isinglass is a high-grade collagen produced from dried air bladder or swim bladder of certain fishes viz. catfish and carps. The processed bladder which is dissolved in hot water forms a gelatin having adhesive property. It is primarily used for clarification of wine, beer and vinegar.

Why are fish so efficient at converting feed to flesh?

13.7.2 Prawn Culture

Most important aquatic crustacean is prawn, which is widely cultured prawn flesh is palatable and rich in glycogen, protein with low fat content.

Types of prawn fishery

1. **Shallow water prawn fishery** – located on the west coast restricted to shallow waters.
2. **Estuaries and back waters or saline lake prawn fishery** - The area of production of prawns are the back waters seen along the Western coast, Ennur, Pulicat, Chilka lake and Estuaries of Ganga and Brahmaputra rivers.
3. **Freshwater prawn fishery** - Prawns are caught from the rivers and lakes throughout India.
4. **Marine prawn fishery** – Most of the marine prawns are caught along the Indian coast belonging to the family Penaeidae.

Species of prawn

A number of species of prawn are distributed in water resources such as *Penaeus indicus, Penaeus monodon, Metapenaeus dobsoni and Macrobrachium rosenbergii.*

Culture of freshwater prawn

Macrobrachium rosenbergii (Figure 13.10) is commonly seen in rivers, fields and low-saline estuaries. The prawn collected from ponds, river, and paddy fields are transferred to the tanks which are aerated. For

XI Std Zoology Chapter-13 EM.indd 253 30/12/2022 14:08:33